

Answer ALL questions. ALL working should be shown.

1. Frame $\{B\}$ is initially coincident with frame $\{A\}$. First, $\{B\}$ is rotated about x_B by 30 degrees, and is subsequently rotated about y_B by 45 degrees, followed by a rotation about z_B by 15 degrees.

- (a) Show that the rotation matrix which accomplishes these 3 rotations in the given order is

$${}^A_B R = \begin{bmatrix} 0.6830 & -0.1830 & 0.7071 \\ 0.5657 & 0.7450 & -0.3536 \\ -0.4620 & 0.6414 & 0.6123 \end{bmatrix}$$

[5 marks]

- (b) Calculate the Euler Parameters for the above rotation matrix.

[3 marks]

- (c) Calculate the unit vector and the single axis of rotation (in equivalent angle-axis representation) for the above rotation matrix.

[2 marks]

- (d) The origin of frame $\{B\}$ is then shifted such that it is at ${}^A P_{BORG} = [5 \ 10 \ 2]^T$. Calculate ${}^B_A T$.

[5 marks]

2. The kinematic structure of a 3-link RPP robot is shown in Figure 2.1. The x_0 axis is given, and the configuration as shown in Figure 2.1 is when $\theta_1 = 0^\circ$.

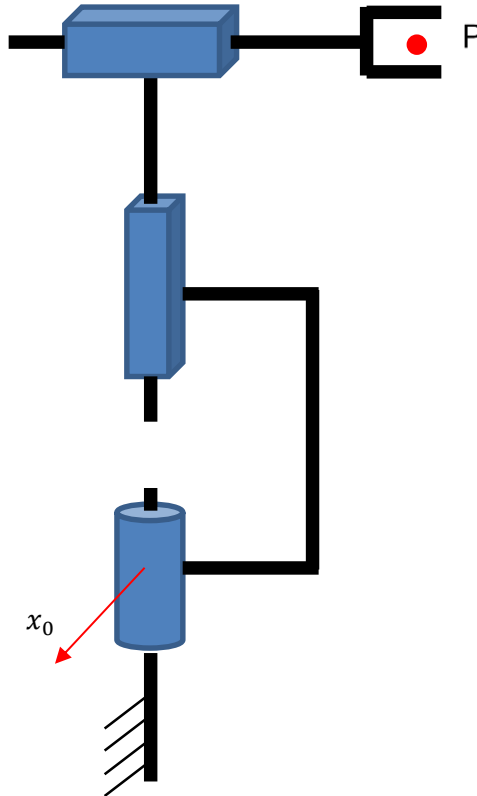


Figure 2.1: RPP robot

- (a) Derive the DH-parameters of the robot, and show that the transformation matrix from frame $\{0\}$ to frame $\{3\}$ is:

$${}^0_3T = \begin{bmatrix} c_1 & 0 & -s_1 & -d_3 \cdot s_1 \\ s_1 & 0 & c_1 & d_3 \cdot c_1 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note:

- If there is any ambiguity in up-down direction, choose up.
- If there is any ambiguity in left-right direction, choose right.
- If there is any ambiguity in into the plane / out of the plane, choose out of the plane.

[10 marks]

- (b) The end-effector (point P) is at a distance of L_3 from the origin of frame $\{3\}$. Derive the linear Jacobian matrix of the end-effector, with respect to the robot's frame $\{0\}$.

[5 marks]

3. An RP robot is shown in Figure 3.1.

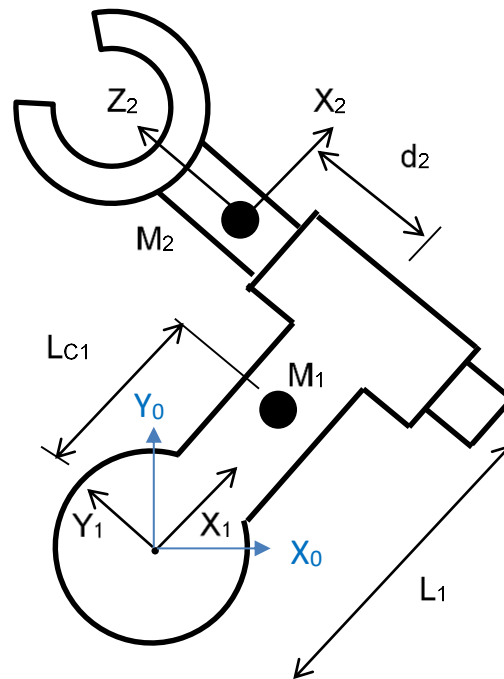


Figure 3.1: RP Robot

Frames $\{0\}$, $\{1\}$ and $\{2\}$, and all the geometrical dimensions of the robot are sketched in Figure 3.1 above. Gravitational pull is in the negative Y_0 direction. M_1 and M_2 are the masses of each link, and the big dots represent the centres of mass. Also, the inertia tensor of the two links with respect to their centres of mass are:

$${}^{c1}I_1 = \begin{bmatrix} I_{XX1} & 0 & 0 \\ 0 & I_{YY1} & 0 \\ 0 & 0 & I_{ZZ1} \end{bmatrix}$$

$${}^{c2}I_2 = \begin{bmatrix} I_{XX2} & 0 & 0 \\ 0 & I_{YY2} & 0 \\ 0 & 0 & I_{ZZ2} \end{bmatrix}$$

The rotation matrices for the robot are as follows:

$${}^0_1R = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

- (a) Perform the outward iteration of the iterative Newton-Euler formula, and show that

$${}^1F_1 = \begin{bmatrix} m_1 s_1 g - m_1 L_{c1} \dot{\theta}_1^2 \\ m_1 c_1 g + m_1 L_{c1} \ddot{\theta}_1 \\ 0 \end{bmatrix}$$

$${}^1N_1 = \begin{bmatrix} 0 \\ 0 \\ I_{ZZ1} \ddot{\theta}_1 \end{bmatrix}$$

$${}^2F_2 = \begin{bmatrix} m_2 s_1 g - 2m_2 \dot{\theta}_1 \dot{d}_2 - m_2 d_2 \ddot{\theta}_1 - m_2 L_1 \dot{\theta}_1^2 \\ 0 \\ m_2 c_1 g + m_2 \ddot{d}_2 + m_2 L_1 \ddot{\theta}_1 - m_2 d_2 \dot{\theta}_1^2 \end{bmatrix}$$

$${}^2N_2 = \begin{bmatrix} 0 \\ -I_{YY2} \ddot{\theta}_1 \\ 0 \end{bmatrix}$$

[10 marks]

- (b) Perform the inward iteration of the iterative Newton-Euler formula to obtain the dynamic equations of the robot. Write the dynamic equations in the $M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \tau$ form.

[10 marks]

4. A two-link planar RR robot is required to move from $(x = 4 \text{ m}, y = 3 \text{ m})$ to $(x = 7 \text{ m}, y = 1 \text{ m})$ within 5 seconds, following a “linear functions with parabolic blend” trajectory.
- (a) What are the minimum accelerations of the blend region in x -direction and y -direction, so that the linear region will exist? If you can only set the same acceleration for both directions, which value would you choose and why?
[3 marks]
- (b) Assume that you have chosen the acceleration of the blend region to be $1 \text{ m} \cdot \text{s}^{-2}$ for both x -direction and y -direction. Derive the equations of the trajectories in x -direction and y -direction.
[12 marks]
- (c) The joint-space dynamic model of the robot is $M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \tau - \tau_{\text{friction}}$. Derive a controller which will ensure that the robot follows the desired trajectory, with a stiffness of k_{p1} for joint 1 and k_{p2} for joint 2. The robot should be critically damped. Additionally, sketch the overall control structure (block diagram) of the robot, given the xy trajectories.
[5 marks]

5. The black-and-white image of a rectangular object has the pixel values as shown in Table 5.1.
(Note: 1st row represents the x-coordinate of a pixel, 1st column represents the y-coordinate of a pixel. They are not part of the image!):

Table 5.1: Pixel values of a rectangular object.

y↓ x→	1	2	3	4	5	6	7	8	9	10
1	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	1	1	0
4	0	0	0	0	0	1	1	1	1	0
5	0	0	0	1	1	1	1	1	1	0
6	0	1	1	1	1	1	1	0	0	0
7	0	1	1	1	1	0	0	0	0	0
8	0	1	1	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0

- (a) Use the “moment” method to find the centroid (leave your answer in 1 decimal point), and show that the inertia matrix of the object is $J = \begin{bmatrix} 126 & -60 \\ -60 & 46 \end{bmatrix}$.

[8 marks]

- (b) From the inertia matrix, calculate the major radius, minor radius and angle of the object. Let all the intermediate calculations and final answer to be in 1 decimal point. Note: Do not use MATLAB to obtain the answer.

[12 marks]

6. Frame $\{B\}$ is initially coincident with frame $\{A\}$. First, $\{B\}$ is rotated about y_A by β degrees, and is subsequently rotated about x_A by α degrees, followed by a rotation about z_A by γ degrees.

(a) Show that the rotation matrix which accomplishes these 3 rotations in the given order is:

$${}^A_B R = \begin{bmatrix} c\gamma c\beta - s\gamma s\alpha s\beta & -s\gamma c\alpha & c\gamma s\beta + s\gamma s\alpha c\beta \\ s\gamma c\beta + c\gamma s\alpha s\beta & c\gamma c\alpha & s\gamma s\beta - c\gamma s\alpha c\beta \\ -c\alpha s\beta & s\alpha & c\alpha c\beta \end{bmatrix}$$

[3 marks]

(b) If the above rotation matrix has the following numerical values,

$${}^A_B R = \begin{bmatrix} 0.6830 & -0.1830 & 0.7071 \\ 0.5657 & 0.7450 & -0.3536 \\ -0.4620 & 0.6414 & 0.6123 \end{bmatrix},$$

calculate the angles α, β and γ .

[7 marks]